Problem 1

a) In any wave $\vec{B} = \frac{\vec{n} \times \vec{E}}{c}$, where $\vec{n}$ is the direction of its propagation.

For incident wave $\vec{n} = -\hat{x}$ and $B_0 = -\frac{E_0}{c}$

Note that the phase $i(kx - \omega t) = -i x_\alpha k^\alpha$ is a 4-scalar and, thus invariant. E.g. for the incident wave in K'-frame: $-kx - \omega t = -k'x' - \omega' t'$

Electric field of the incident wave in K':

$$E_{0y}' = \frac{E_{0y} - V B_{0z}}{\sqrt{1 - V^2/c^2}} = E_0 \sqrt{\frac{1 + V/c}{1 - V/c}}\, e^{-i(k'x' + \omega' t')}$$

Magnetic field of the incident wave in K:

$$B_{0z}' = \frac{B_{0z} - \frac{V}{c^2} E_{0y}}{\sqrt{1 - V^2/c^2}} = -\frac{B_0}{c} \sqrt{\frac{1 + V/c}{1 - V/c}}\, e^{-i(k'x' + \omega' t')}$$

Since $k^\alpha = (\omega/c, \vec{k})$ is a 4-vector the frequency ω' is found from Lorentz transformation:

$$\frac{\omega'}{c} = \frac{\omega/c - V k_x}{\sqrt{1 - V^2/c^2}} = \frac{\omega/c + V \frac{\omega}{c}}{\sqrt{1 - V^2/c^2}}$$

so that $\omega' = \omega \sqrt{\frac{1 + V/c}{1 - V/c}}$

From the boundary condition

$$E_{0y} + E_{Ry} \Big|_{x'=0} = 0$$ we conclude that

the electric field of the reflected wave is

$$E'_{Ry} = -E_0 \sqrt{\frac{1+v/c}{1-v/c}}\, e^{i(k'x' - \omega' t')}$$

Correspondingly, magnetic field is

$$B'_{Rz} = -\frac{E_0}{c}\sqrt{\frac{1+v/c}{1-v/c}}\, e^{i(k'x' - \omega' t')}$$

The reflected wave has the same frequency (otherwise $E'_y = 0$ would not be satisfied at all times)

Performing inverse Lorentz transformation back to K:

$$E_{Ry} = \frac{E'_{Ry} + vB'_{Rz}}{\sqrt{1-v^2/c^2}} = -E_0 \frac{1+v/c}{1-v/c}\, e^{i\tilde{k}x - i\tilde{\omega}t}$$

$$B_{Rz} = \frac{B'_{Rz} + \frac{v}{c^2}E'_{Ry}}{\sqrt{1-v^2/c^2}} = -\frac{E_0}{c}\frac{1+v/c}{1-v/c}\, e^{i\tilde{k}x - i\tilde{\omega}t}$$

where $\tilde{k} = \tilde{\omega}/c$ and $\frac{\tilde{\omega}}{c} = \frac{\omega'/c + \frac{v}{c}k'}{\sqrt{1-v^2/c^2}} = \frac{\omega'}{c}\sqrt{\frac{1+v/c}{1-v/c}}$

Thus $\tilde{\omega} = \omega \frac{1+v/c}{1-v/c}$.

b) $E'_y = 0\Big|_{x'=0}$ becomes $E_y - vB_z\Big|_{x=vt} = 0$

in frame K.

This condition means that the tangential (y) component of the Lorentz force $q(\vec{E}+\vec{v}\times\vec{B})$ acting on charges inside the ideal mirror vanish.

c) Assuming that the reflected wave has some <u>arbitrary</u> amplitude $\tilde{E}_R$ in K-frame and some unknown frequency $\tilde{\omega}$ we obtain from the boundary condition:

$$\left(E_0+\frac{v}{c}E_0\right)e^{-ikx-i\omega t}+\left(\tilde{E}_R-\frac{v}{c}\tilde{E}_R\right)e^{i\tilde{k}x-i\tilde{\omega}t}\Big|_{x=vt}=0$$

or

$$E_0\left(1+\frac{v}{c}\right)e^{-i\omega(1+v/c)t}+\tilde{E}_R\left(1-\frac{v}{c}\right)e^{-i\tilde{\omega}(1-v/c)t}=0$$

This condition is satisfied for all times if and only if

$$\tilde{\omega}=\omega\,\frac{1+v/c}{1-v/c}$$

and

$$E_R=-E_0\,\frac{1+v/c}{1-v/c},$$

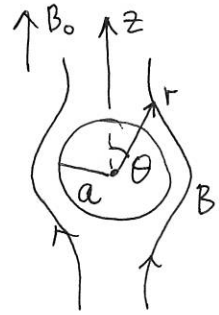
which coincides with the results from part a)

P2

Outside the sphere $\vec{B}(\vec{r}) = -\vec{\nabla}\psi$

$$\nabla^2\psi = 0 \Rightarrow \psi(r,\theta) = -B_0 r\cos\theta + \frac{C_1\cos\theta}{r^2}$$

BC at $r=\infty$: $\psi(r,\theta) = -B_0 z$ since $r\cos\theta = z$.



this satisfies $\vec{B} = -\vec{\nabla}\psi = +B_0\hat{z}$ for $r\to\infty$.

BC at $r=a$: $B_n(r=a) = 0$

In spherical geometry $B_n(r) = B_r(r)$

and $B_r = -\dfrac{\partial\psi}{\partial r} = B_0\cos\theta + \dfrac{2C_1\cos\theta}{r^3}$

$\Rightarrow$ require $\dfrac{2C_1}{a^3} = -B_0 \Rightarrow C_1 = -\frac{1}{2}B_0 a^3$.

Thus $\psi(r,\theta) = -B_0 r\cos\theta - \dfrac{B_0 a^3\cos\theta}{2r^2} =$

$$= -B_0 r\cos\theta - \frac{B_0 a^3 r\cos\theta}{2r^3}$$

$\Rightarrow$ can also write $\psi(x,y,z) = -B_0 z - \dfrac{B_0 a^3 z}{2r^3}$

where $r = \sqrt{x^2+y^2+z^2}$.

Then $\vec{B}(\vec{r}) = -\vec{\nabla}\psi = -\left(\frac{\partial}{\partial x}\hat{x} + \frac{\partial}{\partial y}\hat{y} + \frac{\partial}{\partial z}\hat{z}\right)\left[-B_0 z - \dfrac{B_0 a^3 z}{2r^3}\right] =$

$$= B_0\hat{z} + \frac{B_0 a^3}{2}\vec{\nabla}\frac{z}{r^3} = B_0\hat{z} + \frac{B_0 a^3}{2}\left(\frac{\hat{z}}{r^3} - \frac{3z\vec{r}}{r^5}\right)$$

$\vec{\nabla}\dfrac{z}{r^3} = \dfrac{\hat{z}}{r^3} - \dfrac{3z\vec{r}}{r^5}$ since $\vec{\nabla}\dfrac{1}{r^3} = -\dfrac{3\vec{r}}{r^5}$.

Compare with $\vec{B} = B_0 \hat{z} + \frac{\mu_0}{4\pi}\left(\frac{3(\vec{m}\cdot\vec{r})\vec{r}}{r^5} - \frac{\vec{m}}{r^3}\right)$

Look on coefficient of $\frac{1}{r^3}$ term: $-\frac{\mu_0}{4\pi}\vec{m} = \frac{B_0 a^3}{2}\hat{z}$

$$\Rightarrow \vec{m} = -\frac{2\pi a^3}{\mu_0} B_0 \hat{z} = -\frac{2\pi a^3}{\mu_0}\vec{B}_0 .$$

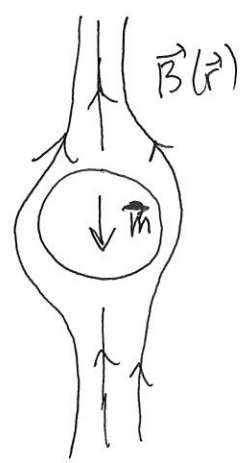
Check now $\frac{1}{r^5}$ term: $\frac{\mu_0}{4\pi} 3(\vec{m}\cdot\vec{r})\vec{r} = \frac{\mu_0}{4\pi}\left(-\frac{2\pi a^3}{\mu_0}\right)\cdot 3 \cdot (\vec{B}_0\cdot\vec{r})\vec{r} =$

$$= -3\left(\frac{1}{2}a^3 B_0\right) z\,\vec{r}$$

which is exactly the coefficient we have.

Thus $\vec{B}(\vec{r}) = B_0\hat{z} + \frac{\mu_0}{4\pi}\frac{3(\vec{m}\cdot\vec{r})\vec{r} - \vec{m}r^2}{r^5}$

with $\vec{m} = -\frac{2\pi a^3}{\mu_0}\vec{B}_0$.



Note that $\vec{m}$ is oriented opposite to external field $\vec{B}_0$: magnetic dipole $\vec{m}$ cancels external field inside superconducting sphere exactly.